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Oxfordian magnetostratigraphy of Britain and its correlation to Tethyan regions and Pacific marine magnetic anomalies

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ABSTRACT

A suite of 11 sections through the Oxfordian (Upper Jurassic) strata in the Dorset and Yorkshire regions of England and the Isle of Skye in Scotland yielded magnetic polarity patterns directly calibrated to the ammonite biostratigraphy of the Boreal and the Subboreal faunal provinces. The sections include the leading candidate for the global stratotype (GSSP) for the Callovian–Oxfordian stage boundary. The mean Oxfordian paleomagnetic pole derived from the Dorset and Yorkshire sections is 71.3°N, 172.6°E ($\delta p = 4.2°$, $\delta m = 6.1°$). The integrated magneto-biostratigraphic scale is consistent with results from the Sub-Mediterranean faunal province and extends the polarity pattern to the base of the Oxfordian. After adjusting for the estimated durations of ammonite subzones from cycle stratigraphy, the magnetostratigraphy confirms models for marine magnetic anomalies M30 through to M37, including some of the short-duration features recorded by deep-tow magnetic surveys in the western Pacific. The Callovian–Oxfordian boundary (base of *Quenstedtoceras mariae* Zone) occurs in a normal-polarity zone that is correlated to the youngest part of polarity chron M37n of this extension to the M-sequence.

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1. Introduction

An integrated time scale of biostratigraphy, magnetostratigraphy and other stratigraphic methods is a powerful tool for precise global correlation. During the late Jurassic through Cenozoic, the M- and C-sequences of marine magnetic anomalies are the reference scale for the nomenclature and the relative durations of magnetic polarity chrons. The combined work of paleomagnetists and stratigraphers have verified this polarity pattern through chron M25 and calibrated it to biostratigraphy (e.g., Berggren et al., 1995; reviewed in Ogg and Smith, 2004).

The oldest magnetic anomaly that is well documented in all ocean basins is M25 of early Kimmeridgian (ca. 150 Ma). The lack of pronounced marine magnetic anomalies in the oceanic crust that formed prior to M25 was once considered to be evidence of an extended Jurassic "Quiet Zone", similar to the long normal-polarity Cretaceous quiet zone. However, detailed surface and deep-tow magnetic surveys over this pre-M25 oceanic crust in the Pacific indicate a complex signature of over 90 close-spaced magnetic anomalies that formed during the Bajocian through Oxfordian of Middle to early Late Jurassic (e.g., Sager et al., 1998; Tivey et al., 2006; Tominaga et al., 2008). In contrast to the well-resolved major magnetic anomalies younger than M25, it is uncertain how many of these short-duration, low-amplitude magnetic anomalies in the deep-tow data represent true geomagnetic polarity chrons and how many are records of paleomagnetic intensity fluctuations.

Our Oxfordian magnetostratigraphy program in Great Britain had five major objectives: (1) Compile a composite magnetic polarity pattern from classic Oxfordian sections in Britain with well-defined Boreal and Subboreal ammonite subzones. Incorporating three independent regions with different facies successions (Dorset, Yorkshire, and Isle of Skye) enabled verification of the main polarity signatures. (2) Incorporate approximate correlations of the ammonite subzones of Boreal and Subboreal realms (temperate to high-latitude fauna) to their Sub-Mediterranean counterparts in the Tethyan realm (tropical fauna) to merge the British-derived polarity composite and the independent Tethyan-derived magnetic polarity zonation (e.g. Przybylski and Ogg, 2007; Przybylski, 2008; Przybylski et al., in press-a,b). This exercise also yielded a more detailed correlation of ammonite zonal schemes between these two realms. (3) Adjust the combined Oxfordian magnetic polarity pattern for the durations of ammonite subzones derived from cyclostratigraphic scaling (e.g. Boulila et al., 2008, in press; Boulila, 2008) to produce magnetic polarity time scale. (4) Correlate the main features of the unified British and Tethyan polarity time scale to the pre-M25 marine magnetic anomaly sequence from deep-tow surveys. This projects precise geologic ages onto the pre-M25 Jurassic crust and enables calibration of the Pacific spreading rates during Oxfordian.

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(5) Analyze and correlate the magnetostratigraphy of candidates for the Global Stratotype Section and Point (GSSP) for the international Callovian–Oxfordian boundary. This provides a secondary high-resolution global-correlation tool for this future GSSP and quantifies the temporal distribution of other correlation horizons.

The uppermost Oxfordian and lowermost Kimmeridgian magnetostratigraphy of four other sections in Great Britain are detailed in a companion report on the correlation of the Oxfordian–Kimmeridgian boundary among faunal realms (Przybylski, 2008; Przybylski et al., in press-a).

2. Oxfordian magnetostratigraphy from the Sub-Mediterranean province and models of Oxfordian marine magnetic anomalies

Magnetostratigraphic sections in condensed pelagic strata of southern Spain indicated that the Oxfordian–Kimmeridgian boundary, as traditionally placed within the Sub-Mediterranean faunal province, was near polarity zone M25r (Ogg et al., 1984), and expanded sections from Crussol, France have indicated placement near the base of this chron (Przybylski et al., in press-a). A portion of the middle to late Oxfordian (upper *Gregoryceras transversarium*, *Perisphinctes bifurcatus*, and lower *Epipeltoceras bimammatum* ammonite zones) in Spain yielded a reproducible magnetic polarity scale (Steiner et al., 1986; verified by Juárez et al., 1994 and Juárez et al., 1995). Ogg and Gutowski (1996) compiled these studies and several additional ammonite-zoned Oxfordian magnetostratigraphic sections from Poland (including sections of Ogg et al., 1991) and Spain into a schematic Oxfordian magnetic polarity time scale for the Sub-Mediterranean ammonite zones. In their proposed suite of correlations to the main features of an early pre-M25 magnetic anomaly model (Handschumacher et al., 1988), the top of the *Quenstedtoceras mariae* Zone (lowest ammonite zone of Oxfordian) was tentatively correlated with marine magnetic anomaly M35n (= M34n in revised numbering system of Sager et al., 1998; Tivey et al., 2006). However, this Sub-Mediterranean polarity scale and its suggested correlations suffered from limited magnetostratigraphic data from several intervals of the Oxfordian and by partial dissimilarity with the Handschumacher et al. (1988) marine magnetic anomaly pattern. Other magnetostratigraphic studies of Oxfordian strata observed the presence of frequent polarity reversals, but the limited biostratigraphic control (e.g., Channell et al., 1990) or uncertainties in deciphering polarity zones in borehole logs (Bouisset and Augustin, 1993) precluded incorporation into a detailed magneto-biostratigraphic time scale. Preliminary magnetostratigraphic studies of proposed boundary stratotypes for the Callovian–Oxfordian interval in the French Alps (near Serres) yielded only normal polarity (Ogg, Atrops and Meléndez, unpublished), but this was an apparent artifact of a persistent secondary overprint and widespread resetting of the magnetization of Mesozoic strata in SE France during Alpine tectonics (Kechra et al., 2003).

These limitations were partly resolved when a nearly complete polarity pattern spanning the middle and upper Oxfordian in the Sub-Mediterranean province has compiled from over two dozen Oxfordian magnetostratigraphic sections and composites (Przybylski, 2008; Przybylski et al., in press-a,b). Even though the lower Oxfordian was poorly resolved due to discontinuities or condensations in the Spanish and Polish reference sections, this polarity scale from the Tethyan faunal province provides the complement to our scale from the Boreal faunal provinces.

Independently, the magnetic anomaly signature of Callovian–Oxfordian-aged crust in the Pacific was refined by a series of deep-tow surveys (Sager et al., 1998; Tivey et al., 2006; Tominaga et al., 2008) that resolved and extended earlier interpretations of pre-M28 magnetic anomalies (Cande et al., 1978; Handschumacher and Gettrust, 1985; Handschumacher et al., 1988; Nakanishi et al., 1992; Channell et al., 1995). These surveys extended the Japanese magnetic lineation pattern produced at the Pacific–Izanagi spreading center to "M45r" (Tominaga et al., 2008). This deep-tow survey significantly revised the details of the previous Handschumacher et al.'s (1988) scale, in addition to producing an offset of the pre-M29 numbering (i.e., "M30" of Sager et al., 1998, is equivalent to "M31" of Handschumacher et al., 1988). Unfortunately, the marine magnetic anomaly pattern from M25r through M27n (Cande et al., 1978) has not yet been enhanced by deep-tow or other surveys. The pre-M25 anomaly ages are constrained by ODP Site 765 on anomaly M26n (155.3 ± 3.4 Ma by Ludden (1992) and near Oxfordian–Kimmeridgian boundary) and ODP Site 801 on anomaly M42n (167.4 ± 1.4 Ma by Koppers et al., 2003; which is possibly Bajocian). However, Sager et al. (1998) suggest that much of the apparent fine-structure of the deep-tow magnetic model probably include some paleointensity variations misinterpreted as polarity chrons, because "unfortunately, any magnetic anomaly data are nonunique, and ultimately, verification whether a wiggle is a wobble or a flip must come from magnetostratigraphic confirmation". Therefore, derivation of the actual geomagnetic time scale for this interval awaits compilation and verification from magnetostratigraphy studies, such as the one in this study.

3. Oxfordian sections of Britain

During the Oxfordian, an intricate system of interconnected, epicontinental marine basins covered northwest Europe, with Britain spanning the transition between the tropical-latitude Sub-Mediterranean province and the high-latitude Boreal and Subboreal provinces. We sampled the Oxfordian interval in three British basins (Fig. 1; Table 1): coastal sections in Dorset representing the Wessex Basin, coastal and quarry sections in Yorkshire–Lincolnshire representing the Cleveland Basin, and coastal sections on Isle of Skye representing the Hebrides Basin.

All sections have well-documented ammonite biostratigraphy (Subboreal province in Dorset and Yorkshire; Boreal to Subboreal province transition in Isle of Skye) and record common sequence stratigraphic packages produced by at least regional global sea-level oscillations (e.g., Coe, 1992, 1995; Wright, 2001a). These sequences are superimposed on a long-term regressive–transgressive cycle represented by offshore marine mudstone-rich facies of the lower Oxfordian (e.g., Weymouth Member of the Oxford Clay Formation) which grade up into the shallow-marine calcareous and coarse-grained siliclastic facies of the middle Oxfordian Corallian Group and are overlain by upper Oxfordian–early Kimmeridgian offshore marine mudstone facies (e.g., Ampthill Clay and Kimmeridge Clay Formation). Maximum regression is interpreted to be near the top of the *Perisphinctes parandieri* ammonite Subzone of the Subboreal province and the *Cardioceras tenuiserratum* Subzone of the Boreal province. [*Note: The term "mudstone" is used in the British sense to encompass a mixture of clay- and silt-sized particles. The informal terms of "lower, middle, upper" Oxfordian are used here, because the formal terms of "Upper, Middle, Lower" used in the literature refer to different intervals depending on the regional ammonite zonal scheme.*]

Samples were collected at 0.5 to 1 m intervals, with denser sampling at the stage boundaries, within condensed intervals and close to stratigraphic hiatuses. Stratigraphic diagrams of each section with the ammonite zones, lithostratigraphy and magnetostratigraphic samples are included in Figs. 3–6 and in Suppl. Figs. 2–6.

3.1. Dorset (Wessex Basin)

The Dorset coast exposes a ~120-m-thick section of marine facies representing most of the Oxfordian (Coe, 1995). A general eastward regional dip is superimposed on minor folding and faulting episodes of pre-Aptian, Aptian and Miocene age.

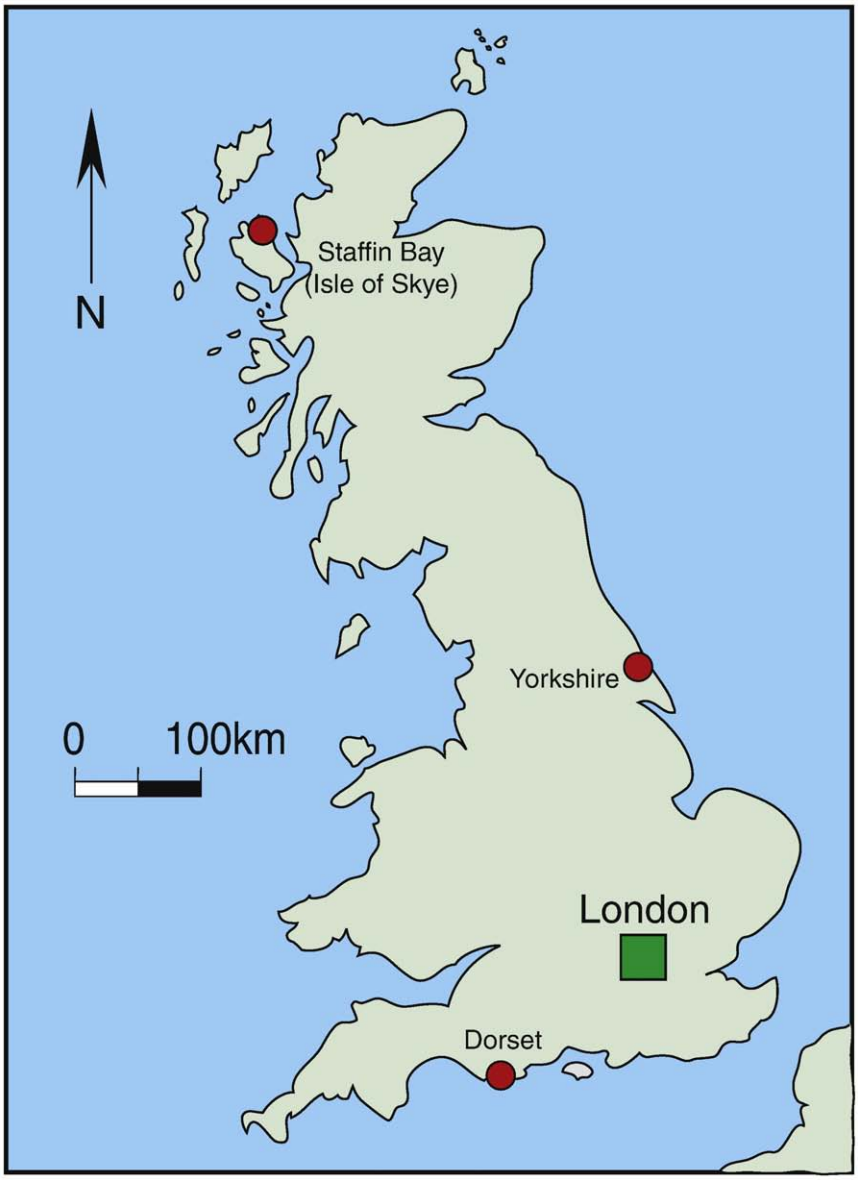


Fig. 1. Geographical location of the clusters of magnetostratigraphy outcrops in Dorset, Yorkshire and Isle of Skye. See Table 1 for further details on individual sections.

The upper part of the *Quenstedtoceras lamberti* Subzone and lower part of the *Q. mariae* Zone appear to be continuous within medium-gray mudstone exposed in a small horst 200 m east of Ham Cliff (Callomon and Cope, 1995; Page, 2004; Page et al., 2006; see Table 1). This section (our *HCO*) is the leading candidate for the global stratotype of the Callovian–Oxfordian boundary (Page et al., 2006). Nearby is Redcliff Point (*RP*), where a small upfaulted block of Jordan Cliff Clay Member (*Cardioceras bukowskii* Subzone) is overlain by Bowleaze Clay Member (*Cardioceras costicardia* Subzone (Wright, 1986b, 2001b)). Furzy Cliff (*FC*) exposes similar Jordan Cliff Clay overlain by Bowleaze Clay, with its distinctive sideritic Red Nodule Bed. Exact relative placement of the sampling sites within each subzone at these coeval sections is uncertain, but their relative positions have been estimated assuming a minimum amount of displacement on the east–west striking fault near the Point (A. Coe and J. Callomon, unpublished field notes 1997).

The array of exposures at Furzy Cliff, Redcliff (*RC*), west Ham Cliff (*HC*), Bran Point (*BP*), Ringstead Bay and Black Head (*BH*) span the *C. bukowskii* Subzone to the *Ringsteadia evoluta* Subzone of the Oxfordian and the *Pictonia baylei* and *Rasenia cymodoce* zones of the Kimmeridgian. The sedimentology and sequence stratigraphy interpretations are described in Coe (1992, 1995). Ammonite zonation is based on Wright (1986a,b, 1997a,b; see Table 1).

A stratigraphic overlap of the sections at Black Head and Bran Point provides a fold test. Strata dip 45° to the north at Black Head, but are relatively horizontal at Bran Point. Note that the unit named Nodular Rubble Member in this interval is a misnomer applied to the striking bioturbation of the *P. parandieri* Subzone, rather than actual clasts.

3.2. Yorkshire–Lincolnshire (Cleveland Basin and East Midlands Shelf)

The Yorkshire coast and inland quarries expose strata deposited in the Cleveland Basin and adjacent East Midland shelf of the North Sea basin system (Rawson and Wright, 1995). The region was uplifted into a gentle E–W oriented anticline during latest Cretaceous or Paleogene.

Cunstone Nab (*CN*) had been considered as a candidate for the Callovian–Oxfordian GSSP, but it is very condensed (Page, 2004). Sample suites spanning the lower and middle Oxfordian were collected at Filey Brigg (*FB*) and Newbridge Quarry (*NQ*) (Wright, 1972, 1983; Coe, 1995; Wright, 1996, 1997b).

Table 1
Summary information for Oxfordian sampling sites — location, position, stratigraphic interval, average bedding orientation, number of samples (a total of 282 minicores were analyzed), year of sampling and selected references for lithostratigraphy and biostratigraphy. Abbreviations for ammonite genera: "R." = *Ringsteadia*, "P." = *Perisphinctes*, "C." = *Cardioceras*, "Q." = *Quenstedtoceras*. Color shadings are to highlight regional groupings of sections.

Section	Location	Approximate position		Biostratigraphic span of sampling	OS grid reference	Average bedding		Analyzed samples	References (litho- and biostrat)
		Lat (°)	Lon (°)			Dip (°)	Dip Dir (°)		
Black Head (BH)	Dorset	50.636°N	2.386°W	*R. pseudocordata – P. pumilus*	SY726819	46	11	37	Coe 1992, (1995); Wright 1986a, (1997a, 2001a)
Bran Point (BP)	Dorset	50.630°N	2.365°W	*P. cautisnigrae – P. plicatilis*	SY742813	6	73	33	Coe 1992, (1995); Wright 1986a, (1997b, 2001a)
West Ham Cliff (HC)	Dorset	50.633°N	2.407°W	*P. plicatilis*	SY712816	5	210	7	Coe 1992, (1995); Wright 1997b, (2001a)
Redcliff (RC)	Dorset	50.635°N	2.416°W	*P. plicatilis*	SY706818	7	10	8	Coe 1992, (1995); Wright 1997b, (2001a)
Furzy Cliff (FC)	Dorset	50.633°N	2.427°W	*C. cordatum*	SY698818	3	90	9	Coe 1992, (1995); Wright 1986b, (2001a)
Redcliff Point (RP)	Dorset	50.632°N	2.407°W	*C. cordatum*	SY712816	2	300	10	Coe 1992, (1995); Wright 1986b, (2001a, 2001b)
East Ham Cliff (HCO)	Dorset	50.634°N	2.406°W	*Q. mariae – Q. lamberti*	SY715818	29	358	30	Coe 1992, (1995); Callomon and Cope, (1995); Page 2004; Page et al., (2006); Wright (2001a)
Newbridge Quarry (NQ)	Yorkshire	54.262°N	0.775°W	*P. pumilus*	SE800860	3	225	15	Coe (1995); Wright (1972, 1996)
Filey Brigg (FB)	Yorkshire	54.217°N	0.265°W	*P. plicatilis – Q. mariae*	TA126817–TA139814	3	218	31	Coe (1995); Wright (1972, 1983, 1997b)
Cunstone Nab (CN)	Yorkshire	54.232°N	0.312°W	*Q. mariae – Q. lamberti*	TA100831	5	185	12	Page (2004); Wright (1983)
Staffin Bay (lower part)	Isle of Skye, UK	57.651°N	6.236°W	*C. tenuiserratum – Q. lamberti*	NG473698–NG472708	40	293	90	Cox (1990); Page (2004); Riding and Thomas, (1997); Sykes and Callomon, (1979)
							Total =	282	

3.3. Isle of Skye (Hebrides Basin)

Oxfordian marine mudrocks were sampled at Staffin Bay along the NE coast of the Isle of Skye (Suppl. Fig. 2). These clay-rich strata are located in the intertidal zone in a series of small fault blocks and landslips. Due to their richness of ammonite assemblages and continuity of deposition, these outcrops are significant for the Boreal and Subboreal zonations (Sykes and Callomon, 1979; Wright, 1989; Cox, 1990; see Table 1) and are candidates for the global boundary stratotypes of the Callovian–Oxfordian and the Oxfordian–Kimmeridgian boundaries (Page, 2004; Wierzbowski et al., 2006; Matyja et al., 2006). The magneto-biostratigraphy across the Oxfordian–Kimmeridgian boundary interval was instrumental in correlating from this Boreal realm to the Tethyan realm (Przybylski, 2008; Przybylski et al., in press-a). The abundant dinoflagellate cysts within the gray mudstones are calibrated to Boreal and Subboreal ammonite zones (Riding and Thomas, 1997) but the resolution is poor. The high-resolution Oxfordian ammonite stratigraphy and distinctive beds at Point 3 and Point 5 of Sykes and Callomon (1979) were used to construct a composite section. The thick succession of ammonite-zoned Callovian strata underlying the *Q. lamberti* Zone on the Isle of Skye also yielded both normal- and reverse-polarity intervals, but these preliminary results await verification.

4. Paleomagnetic analytical procedure and magnetic properties of the Oxfordian samples from Britain

4.1. Demagnetization procedure and magnetic behavior

The majority of the 282 oriented minicores (Table 1) were measured at the cryogenic magnetometer facility at the University of Michigan. The suites from east of Ham Cliff (HCO) were measured at Oxford University. Progressive thermal demagnetizations were tailored to each exposure based on pilot studies of each lithology from each region, but generally consisted of 30 °C steps within the typical range that yielded characteristic directions. Bulk susceptibility was generally monitored after every second thermal demagnetization step below 300 °C and after every temperature step above 300 °C. Fissile mudstone minicores were wrapped in aluminum foil to prevent disintegration during analysis. Such wrappings had no measurable magnetization in the magnetically shielded room at the University of Michigan; however, they did acquire spurious magnetizations within the Helmholtz-coil field-free spaces at the University of Oxford, so this suite of friable samples required re-assembly with non-magnetic glue after each heating step.

Even though magnetic characteristics varied greatly among the various lithologies at each outcrop, there were some general characteristics. Initial NRM directions are typically dominated by normal polarity oriented parallel with the present dipole field. This secondary overprint was largely removed upon heating above 200 °C, then divergent polarities among samples were typically observed through the demagnetization process from 250 °C through 350 °C. Upon heating above 350 °C, most samples displayed a rapid increase in magnetic susceptibility accompanied by viscous magnetizations. Despite performing the thermal demagnetization and measurements within a shielded room, the onset of spurious viscous magnetizations generally rendered measurements beyond 400 °C to be random noise.

Gray mudstones and the few samples of sideritic limestone typically had much higher bulk susceptibilities than the limestones and siliclastic sandstones. A few of the mudstones displayed anomalous paramagnetic behavior, in which the weak magnetic field within the shielded room would be instantly and continuously acquired by the minicore just before it entered the cryogenic magnetometer. This paramagnetic behavior was mainly restricted to a few mudstones from the Isle of Skye, but rendered even their initial NRM directions to be merely random lab magnetizations.

The typical magnetic behavior during thermal demagnetization suggests that a common carrier of characteristic magnetizations

(generally the 200 °C–400 °C range) is magnetite. The surge in susceptibility above 350 °C observed in many samples is probably due to a combination of dehydration of iron-rich clays and the oxidation of iron sulfides. In the few organic-rich mudstones, combustion during thermal demagnetization may have aided alteration of other iron minerals.

4.2. Interpretation of polarity and calculation of paleomagnetic directions

Polarity interpretations of the magnetization components unblocked in the 200–400 °C range were generally obvious from the characteristic directions. Most samples either attained a direction which remained stable through multiple heating steps as the intensity of magnetization decreased (i.e., univectoral decay to the origin on the vector diagrams), or displayed development of a semi-stable direction and intensity prior to onset of erratic magnetization at higher temperatures. However, for a few samples, especially the mudstone minicores from the Isle of Skye, a stable end-point characteristic direction was not achieved prior to onset of viscous magnetization. For these samples, it was necessary to deduce their polarity from their progressive demagnetization trends along great-circle paths toward the typical "normal" or "reversed" directions derived from less-viscous samples in the outcrop (see examples in Wierzbowski et al., 2006). Characteristic magnetization directions and associated variances were computed for each sample by applying a least-squares three-dimensional line-fit (procedure of Kirschvink, 1980) to sets of vectors displaying removal of a single component in equal-area and vector plots of the progressive demagnetization. The intensity of characteristic magnetization was computed as the mean of the intensities of those vectors used in the least-squares fit.

Each characteristic direction was assigned a polarity rating based upon the individual demagnetization behavior: (1) N or R implies well-defined directions computed from at least 3 vectors, (2) less-precise NP or RP have directions computed from only 2 vectors or were samples displaying high dispersion, (3) NPP or RPP-rated samples had obvious polarity trends, but did not achieve adequate cleaning during demagnetization; these were omitted from mean pole computations, (4) and samples with uncertain (N? or R?) or indeterminate (INT) polarity that were not used to define polarity zones. Typical demagnetization behavior associated with each of these ratings is displayed in Fig. 2A–F.

Polarity interpretations were done independently by two paleomagnetists (by J.G.O. with A.L.C, and by P.A.P.) to reduce the bias of a single observer. The more conservative of these two independent assignments was generally used.

5. Paleomagnetic poles

For each section and for each regional composite, mean directions of the characteristic magnetizations and corresponding apparent poles calculated separately for each polarity using only the high-rated samples (those with N and R ratings; plus NP- and RP-rated samples given half weighting) with a method modified from Fisher (1953). The intensity of each mean direction vector was computed as the logarithmic average of the incorporated characteristic direction intensities. After the first iteration, sample directions falling outside two standard deviations from the initial mean were omitted. The mean characteristic direction for each outcrop is generally the combined results of normal- and reverse-polarity sets, but a vector-sum method that incorporates the mean intensities of the normal- and reversed-polarity vectors was also employed for comparison. In some cases, especially when the relative numbers of normal- and reversed-polarity samples differ significantly from each other and there may be a small residual overprint on all samples, the sum of the mean normal-polarity vector and the reversed-polarity vector can yield a less distorted mean characteristic direction. A similar suite of computations was used for regional composites (Suppl. Fig. 1A).

In nearly all exposures, the mean normal-polarity and reversed-polarity characteristic directions are antipodal to within 20°. The significance of these normal- and reversed-polarity directions was partially tested by the reversal tests of Butler (1998) and McFadden and McElhinny (1990). However, because these tests ideally require more than five highest-quality samples of each polarity and most of our sections do not provide this minimum, we merged the sections from Dorset and from Yorkshire into two regional composites (Table 2A). The mean British Oxfordian paleomagnetic pole derived from Dorset and Yorkshire composites passed both significance tests and received a class B rating on the McFadden and McElhinny test.

The suites of mean directions from sections and regional composites display several features:

(1) The mean normal- and reversed-polarity directions of each regional composite (Table 2A) are within 10° of antipodal (alpha-95 typically less than 6°). The low value (30°) of the precision parameter (K) reflects the high dispersion of the component directions. The characteristic directions of magnetization of these British Oxfordian sets are distinct from present-day magnetization.

(2) The Isle of Skye sections display an apparent clockwise rotation of approximately 30° relative to the composite Dorset–Yorkshire mean declinations. Even though these sections are within landslipped blocks, the suites display similar declinations regardless of their structural corrections (the field dips ranged from 30° to nearly vertical). We suggest that part of this offset from Dorset–Yorkshire declinations is a post-Jurassic clockwise rotation of this Staffin region of the Isle of Skye relative to the main part of Britain, and omitted these sections from the calculation of a mean British Oxfordian paleopole. These Isle of Skye mudstones yield anomalously shallow paleomagnetic inclinations with an apparent mean paleolatitude of only 27°N (Suppl. Table 1); which is undoubtedly an artifact of inclination shallowing within these clay-rich facies.

(3) In regions where adjacent exposures had different bedding orientations (Isle of Skye and Dorset), the mean directions of the sites within each region statistically coincide after applying structural corrections (Suppl. Fig. 1B; Suppl. Table 1). For example, the mean directions from Bran Point (BP) and Black Head (BH) are dispersed by 50° before tilt correction, but are within 10° of each other after structural correction.

(4) The mean Oxfordian direction from Dorset (2° declination, 50° inclination; ~6° alpha-95; Table 2, "Vector comb.") is rotated clockwise by about 12° from the tilt-corrected mean direction obtained from the steep-dipping Purbeck Limestone Formation of uppermost Jurassic and lowest Cretaceous (Ogg et al., 1994). Portland Beds on the Isle of Portland yielded a mean direction with the same inclination, but with a mean declination rotated clockwise by 34° relative to the main Dorset coastal sections (Hasenyager, 1990), which suggests local block rotations.

(5) The Dorset sites yield a mean paleolatitude of 31°N (about ±5°). The mean paleolatitude of the Yorkshire sites, which are presently 4° further north, is 37°N (about ±7°) (Table 2A).

(6) The mean British Oxfordian paleopole, as computed from seven Dorset and three Yorkshire sections is 71.3°N, 172.6°E ($\delta p = 4.2°$, $\delta m = 6.1°$) (Table 2A). This is within 8° to a nearly coeval Oxfordian mean pole from Poland (Przybylski, 2008; Przybylski et al., in press-b) and is within 6° to a mean Oxfordian–Kimmeridgian boundary paleopole for the European Craton (Przybylski, 2008; Przybylski et al., in press-a) (Table 2B). However, that Oxfordian–Kimmeridgian boundary paleopole incorporated three British sites (Dorset's upward continuation of Black Head section and the nearby Ringstead

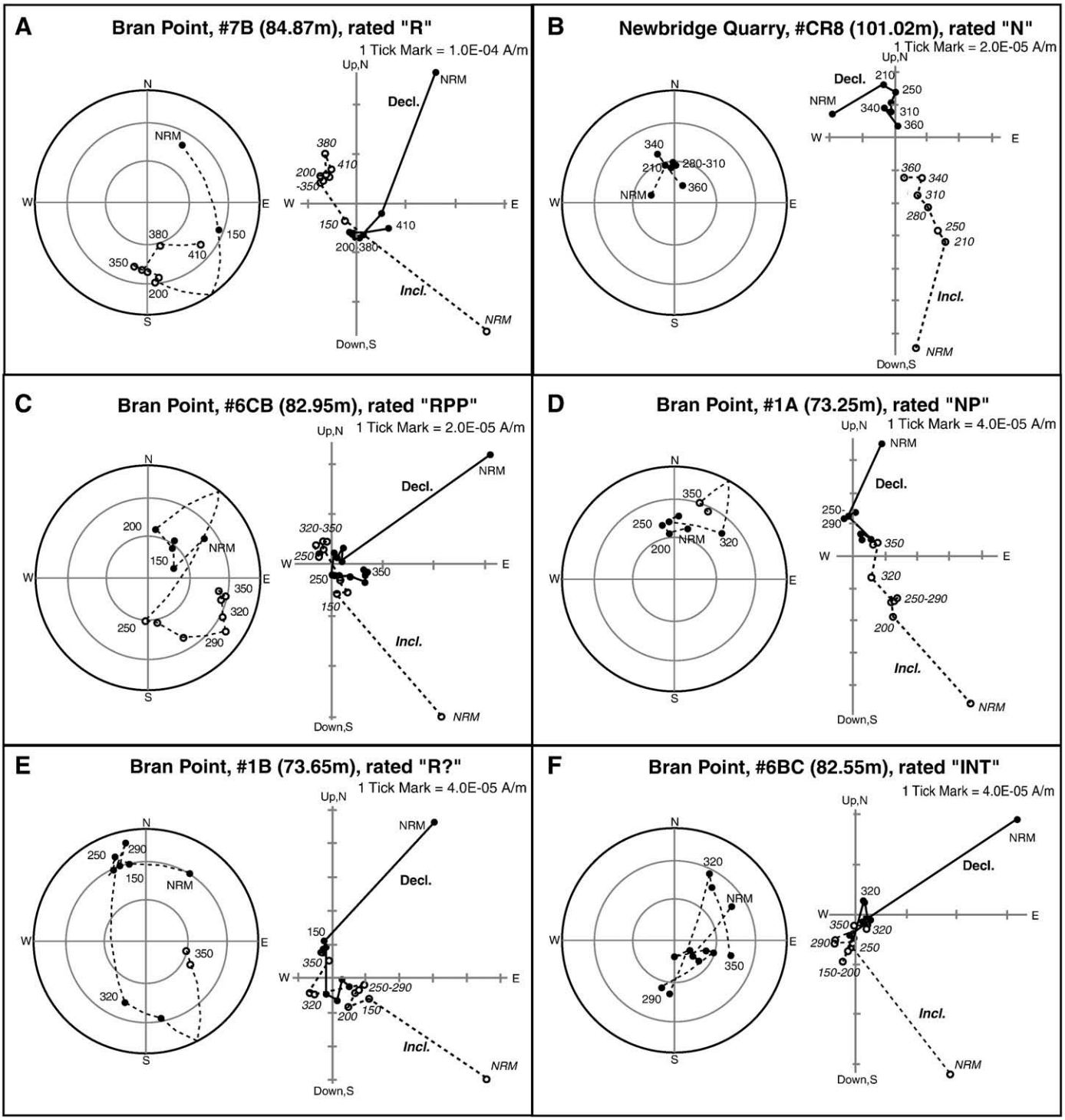


Fig. 2. Stereographic projections (left) and Zijderveld diagrams (right) of typical thermal demagnetization behavior of samples of different polarities (A–F). See explanation in text. Declination and inclination vectors (solid lines and dots versus dashed lines with open dots) correspond to the horizontal component and to the actual dip of the magnetization vector; vectors have been corrected for tilt of strata. A. Sample BP_7B, a medium-gray wackestone–packstone from the Bran Point quarry is rated "R" based upon its stable endpoint. Onset of surge in susceptibility and viscous behavior at 380 °C; characteristic vector obtained from 4 thermal steps, 250°–350° = 184.6° declination, −39.9° inclination, 8.7×10^{-5} A/m intensity. B. Sample NQ_CR8 taken from the gray limestone at Newbridge Quarry is rated "N" based upon its tight clustering and univectoral decay; characteristic vector obtained from 3 thermal steps, 250°–310° = 358.5° declination, 64.8° inclination, 5.0×10^{-5} A/m intensity. C. Sample BP_6CB (light yellowish-gray oolitic limestone), characteristic vector (4 measurements, 250°–290° = 149.8° declination, −44.6° inclination, 7.8×10^{-6} A/m) is rated "RPP" based upon weak intensity and associated high error. Shift at 320–350 °C may be drift toward overprint. Definite reversed polarity, but not useful for computing mean poles. D. Sample BP_1A (siltstone with brown clay streaks), characteristic vector (3 thermal steps, 250°–290° = 355.3° declination, 46.2° inclination, 7.5×10^{-5} A/m) is rated "NP" based on its stable normal trend but very small linear decay. E. Sample BP_1B (light brownish-gray fine-grained limestone), characteristic vector (2 measurements at 350° = 113.4° declination, −55.1° inclination, 3.0×10^{-5} A/m) is rated "R?" based upon its apparent flip to reversed-polarity hemisphere at higher temperatures but failure to achieve a stable endpoint. Uncertain polarity (not used for assigning zones) and omitted from pole computations. F. Sample BP_6BC (gray oolitic limestone with tan mottles), characteristic vector (4 measurements, 250°–290° = 163.1° declination, 63.4° inclination, 2.7×10^{-5} A/m) is rated "INT" (indeterminate) based upon total lack of stable directions during demagnetization, although initial drift may indicate reversed polarity (it is adjacent to reversed-polarity samples).

Table 2
(A) Summary of mean directions and poles (with statistical significance tests) for the Dorset suite of sections, for Yorkshire suite, and for the Dorset–Yorkshire composite. Bold font highlights mean directions displayed in Suppl. Fig. 1. (B) Comparison to selected late Jurassic virtual geomagnetic poles from European Craton.

A. British Oxfordian mean directions - combined results

			After bedding correction													Reversal tests	
Location	Section composite	Magnetic directions	Least squares (weighted)							Mean virtual geomagnetic pole						Butler 1998	McFadden and McElhinny (1990)
			Number	Decl (°)	Incl (°)	Intensity (A/m)	α 95 (°)	K	R / N	Lat (°)	Lon (°)	δp (°)	δm (°)	Paleo Lat (°)	Plat (+/-)	F / Ft	Y / Yc
British Oxfordian, Dorset	Composite of 7 sections (HCO, RP, FC, RC, HC, BP, BH)	Normal	15.5	9.0	46.0	3.86E–05	8.1	22.4	14.85/15.5	64.5	159.4	6.6	10.4	27.4		**1.52/3.22**	**9.16/12.80**
		Reverse	8.0	178.3	–51.9	6.89E–05	9.4	35.6	7.80/8.0	–70.6	2.7	8.8	12.9	–32.6			Positive test, class C
		Combined	23.5	5.6	48.2	4.70E–05	6.1	24.7	22.59/23.5	66.9	165.8	5.3	8.0	29.2	+5.6/–4.9	Positive test	
		Vector comb.		2.4	49.9	5.36E–05				**68.7**	**172.6**	5.5	8.2	**30.7**			
British Oxfordian, Yorkshire	Composite of 3 sections (CN, FB, NQ)	Normal	13.5	2.1	60.5	5.24E–05	7.0	34.9	13.14/13.5	76.7	172.5	8.1	10.6	41.5		**0.67/3.32**	**7.16/15.58**
		Reverse	3.5	184.9	–53.5	9.45E–05	19.9	28.5	3.41/3.5	–69.1	348.2	19.3	27.7	–34.0			Test not possible – not enough well-defined reverse directions
		Combined	17.0	2.8	59.1	5.73E–05	6.2	34.3	16.53/17.0	75.0	171.2	6.9	9.2	39.9	+7.4/–6.4	Positive test	
		Vector comb.		4.0	56.0	7.33E–05				**71.6**	**169.3**	6.4	8.9	**36.6**			
British Oxfordian, Dorset and Yorkshire	Composite of 10 sections (HCO, RP, FC, RC, HC, BP, BH, NQ, FB, CN)	Normal	28.5	5.6	53.3	4.47E–05	5.6	24.0	27.35/28.5	71.4	163.6	5.4	7.8	33.8		**0.26/3.13**	**2.80/9.89**
		Reverse	11.5	180.2	–52.4	7.59E–05	7.5	36.1	11.21/11.5	–71.1	357.8	7.1	10.3	–33.0			Positive test, class B
		Combined	40.0	4.0	53.1	5.21E–05	4.4	26.9	38.55/40.0	71.4	167.8	4.3	6.2	33.6	+4.5/–4.0	Positive test	
		Vector comb.		2.2	52.8	6.03E–05				**71.3**	**172.6**	4.2	6.1	**33.4**			

B. Comparison of mean directions and pole position with previous studies

Location	Stratigraphic age	Sites	Lat (°)	Lon (°)	Decl (°)	Incl (°)	Paleo Lat (°)	References
Dorset and Yorkshire, Britain	Oxfordian	10	**71.3**	**172.6**	2.2	52.8	**33.4**	Present study
Southern Poland	Oxfordian	11	**78.5**	**184.9**	358.3	59.7	**40.5**	Przybylski (2008); Przybylski et al. (in press-b)
Polish Jura Chain	Oxf.-Kim. bound.	5	**77.4**	**175.9**	0.7	58.6	**39.4**	Przybylski (2008); Przybylski et al. (in press-a)
European Craton	Oxf.-Kim. bound.	14	**75.0**	**181.2**	359.1	56.4	**37.0**	
Polish Jura Chain	Callovian–Middle Oxfordian	4	**74.3**	**200.3**	352.7	56.6	**37.2**	Ogg et al., (1991)
Dorset, Britain	Tithonian–Berriasian		**71.0**	**208.0**	348.7	54.3	**34.8**	Ogg et al., (1994)
Crussol and Berrias, France	Berriasian		**75.0**	**183.0**	358.5	56.5	**37.0**	Galbrun (1985); Ogg, Atrops and Coe (in prep.)

Abbreviations:

Number, *N*	total number of observations (specimens) used in the analysis (fractional if data is weighted)
Decl	mean geographic declination
Incl	mean geographic inclination
Intensity	mean intensity of the ChRM directions
α 95 [°]	angular radius of the cone of 95% confidence level
K	Fisher precision parameter
R	length of the resultant vector of all individual directions
Lat, Lon	latitude and longitude of the virtual geomagnetic paleopole
δm	radius of the 95% confidence level ellipse of the paleomagnetic pole, measured along the direction of the magnetic meridian from the site to the vertical pole
δp	radius of the 95% confidence level ellipse of the paleomagnetic pole, measured perpendicular to the direction of the magnetic meridian from the site to the vertical pole
PaleoLat	magnetic paleolatitude of the sampling location
+/– on PaleoLat	upper and lower confidence limits on the magnetic paleolatitude; mean values of these limits are given
F	Fisher statistic for Reversal test (Butler 1998)
Ft	tabulated *F* value for $p = 0.05$ for 2 and 2(*N*–2) degrees of freedom
Y	spherical angle between the mean normal and mean inverted reverse direction
Yc	reversal critical angle (McFadden and McElhinny, 1990)
Sites	number of localities

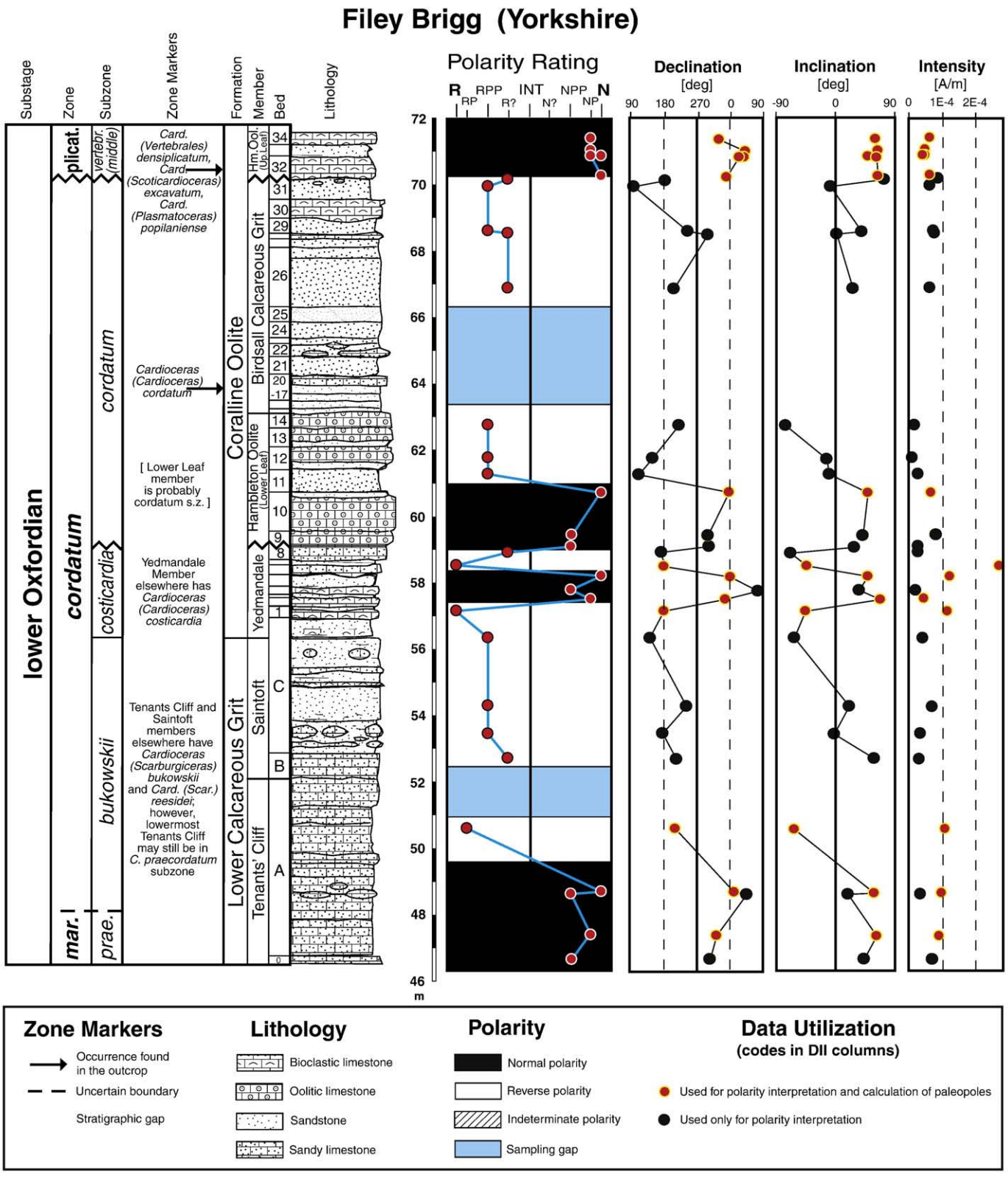


Fig. 3. Filey Brigg (FB, Yorkshire) magnetostratigraphy with placement of main ammonite levels and interpretation of ammonite zonation. The meter scale is according to the regional composite of formation thicknesses. Polarity rating: N/R — good normal- or reversed-polarity behavior, INT — indeterminate or intermediate. Intensity column is for the computed vector for the characteristic direction of each sample. Abbreviations in ammonite columns: "plicat." = *Perisphinctes* (*Liorphinctes*) *plicatilis*, "vertebr." = *Cardioceras* (*Vertebriceras*) *vertebrale*, "Card." = *Cardioceras*, "mar." = *Quenstedtoceras mariae*, "prae." = *Cardioceras praecordatum*. In Member column, "Hm. Ool. (Up. Leaf)" = Hambleton Oolite (Upper Leaf) member.

Bay section and of the South Ferriby section in North Lincolnsire), therefore cannot be used as a test of the stability of Britain relative to the rest of the European plate. The British Oxfordian paleopole is also within 6° of the mean Tithonian–Berriasian pole from Crussol and Berrias in southeastern France (Galbrun, 1985; Przybylski et al., in press-a) (Table 2B).

(7) However, the British-derived Oxfordian poles would place Britain at a slightly more southern paleolatitude than what would be projected from the non-British poles. The non-British poles are mainly derived from limestone-rich sections, whereas the British sections generally incorporate some clay-rich intervals. This suggests that a minor amount of "inclination shallowing" during late-stage compaction of mudstone facies may have produced a slight, but noticeable bias in the mean regional poles of this British suite.

(8) The British array of paleomagnetic sections does not allow resolution of a paleolatitude drift through the Oxfordian. Even though a rapid southward drift of the Mediterranean region (Italy, Western Carpathians) during Callovian–Oxfordian has been proposed (e.g., Muttoni et al., 2005; Lewandowski et al.,

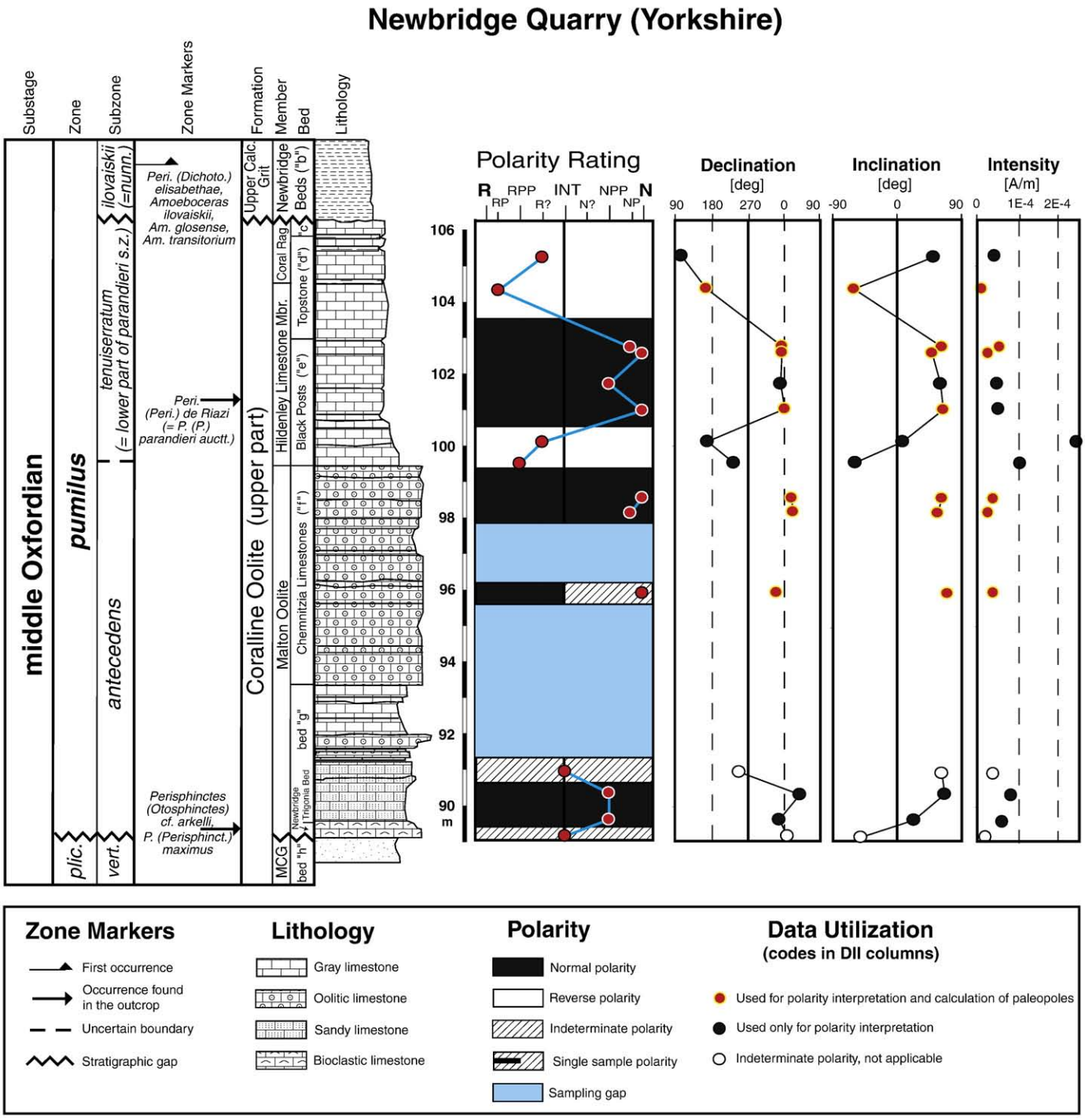


Fig. 4. Newbridge Quarry (NQ, Yorkshire) magnetostratigraphy with placement of main ammonite levels and interpretation of ammonite zonation. The meter scale is according to the regional composite of formation thicknesses. Polarity rating: N/R — good normal- or reversed-polarity behavior, INT — indeterminate or intermediate. Intensity column is for the computed vector for the characteristic direction of each sample. Abbreviations in ammonite columns: "nun." = *Perisphinctes nunningtonense*, "P." or "Peri." = *Perisphinctes*, "Am." = *Amoeboceras*, "plic." = *Perisphinctes* (*Liorphinctes*) *plicatilis*, "vert." = *Cardioceras* (*Vertebriceras*) *vertebrale*.

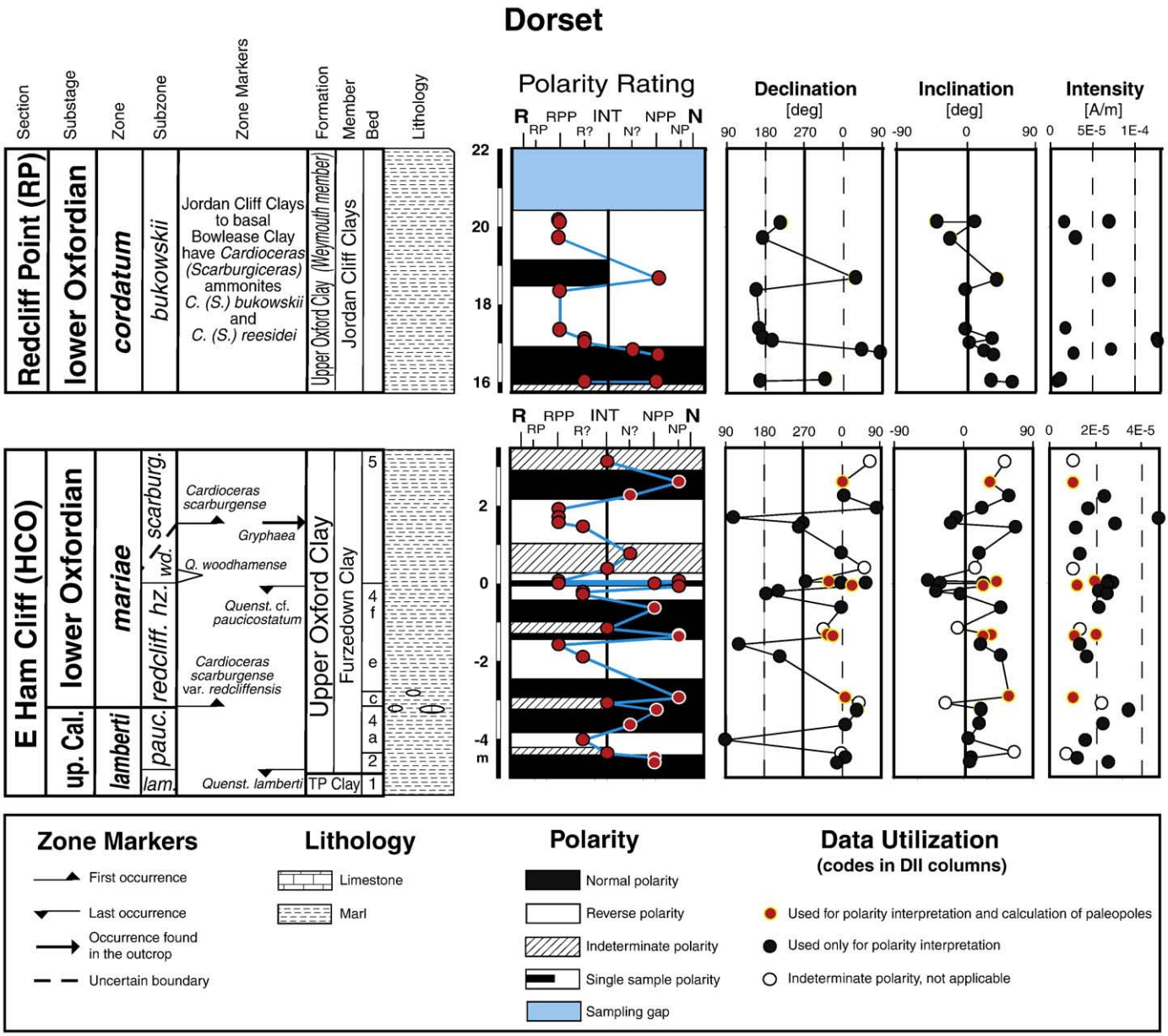


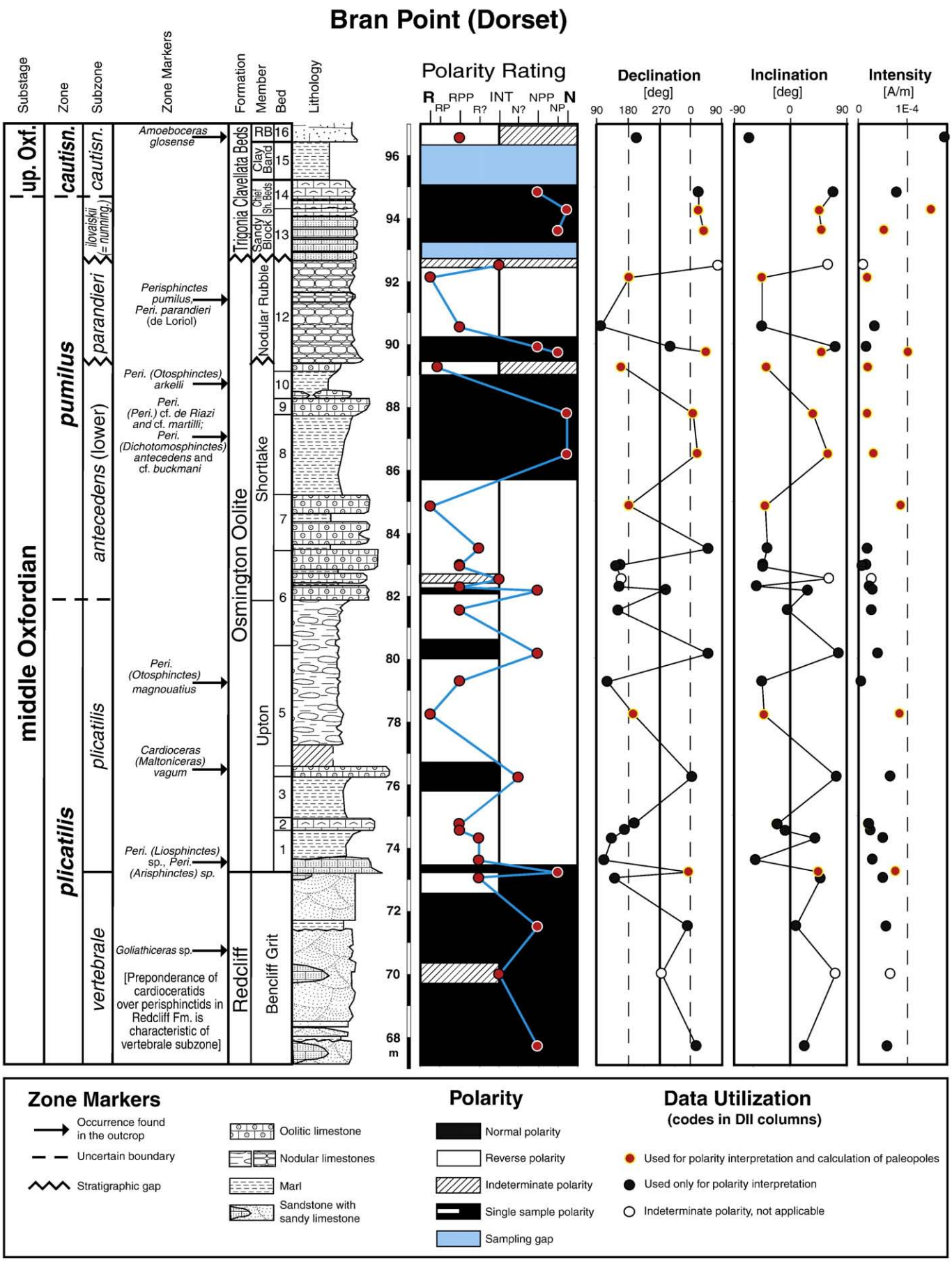
Fig. 5. Magnetostratigraphy of east Ham Cliff (HCO), which is a candidate for the GSSP of the Callovian/Oxfordian boundary, and Redcliff Point (RP), Dorset, with placement of main ammonite levels or range limits and interpretation of ammonite zonation. Section HCO may be within a landslip (S. Hesselbo, pers. commun., Sept 2009). The meter scales are according to the regional composite of formation thicknesses. Polarity rating: N/R — good normal- or reversed-polarity behavior, INT — indeterminate or intermediate. Intensity column is for the computed vector for the characteristic direction of each sample. Abbreviations in ammonite columns: "scarburg." = *Cardioceras* (*Scarburgiceras*) *scarburgense*, "wd." = *Quenstedtoceras woodhamense*, "redcliff. hz." = *Cardioceras scarburgense* var. *redcliffensis* Horizon, "pauc." = *Quenstedtoceras paucicostatum*, "lam." = *Quenstedtoceras lamberti*.

2005), a corresponding motion of the European part of the Laurasia plate has yet to be resolved.

6. Magnetostratigraphic results and a British composite polarity time scale

The suite of sections from Dorset, Yorkshire and the Isle of Skye enables compilation of a consistent composite polarity scale for the Oxfordian of Britain. The main polarity zones, consisting of clusters of samples having similar polarity, are generally well-delimited for each section (Figs. 3–7; Suppl. Figs. 2–6). Ammonite subzones of the Boreal and Subboreal faunal provinces have been inter-correlated and provide the primary means to correlate the main polarity zones. Regional sequence stratigraphy interpretations (e.g., Coe, 1992, 1995) are guides to potential stratigraphic gaps or condensation in the successions. The composite polarity pattern common to all sections (Fig. 7) includes a partial adjustment for the different facies and condensed intervals among regions. Most polarity zones are represented in at least two of these three regions, and there are no significant discrepancies within any biostratigraphic interval. The

Fig. 6. Bran Point (BP, Dorset) magnetostratigraphy with placement of main ammonite levels or range limits and interpretation of ammonite zonation. The meter scale is according to the regional composite of formation thicknesses. Polarity rating: N/R — good normal- or reversed-polarity behavior, INT — indeterminate or intermediate. Intensity column is for the computed vector for the characteristic direction of each sample. Abbreviations in ammonite columns: "cautisn." = *Perisphinctes cautisnigrae*, "nunning." = *Perisphinctes nunningtonense*, "Peri." = *Perisphinctes*. Abbreviations in Member column: "RB" = Red Beds Member, "Chief Sh. Beds" = Chief Shell Beds.


Bran Point (Dorset)
Substage
Zone
Subzone
Zone Markers
Formation
Member
Bed
Lithology
Polarity Rating
R RP RPP INT NPP N
Declination [deg]
Inclination [deg]
Intensity [A/m]
middle Oxfordian
up. Oxf.
cautisn.
pumilus
plicatilis
cautisn.
ilovaiskii (= rumining.)
parandieri
antecedens (lower)
plicatilis
vertebrale
Amoeboceras glosense
Perisphinctes pumilus, Peri. parandieri (de Loriol)
Peri. (Otosphinctes) arkelli
Peri. (Peri.) cf. de Riazi and cf. martilli; Peri. (Dichotomosphinctes) antecedens and cf. buckmani
Peri. (Otosphinctes) magnouatius
Cardioceras (Maltoniceras) vagum
Peri. (Liosphinctes) sp., Peri. (Arisphinctes) sp.
Goliathiceras sp.
[Preponderance of cardioceratids over perisphinctids in Redcliff Fm. is characteristic of vertebrale subzone]
Trigonia Clavellata Beds
Osmington Oolite
Redcliff
RB
Clay Band
Chiel Sh. Beds
Sandy Block
Nodular Rubble
Shortlake
Upton
Bencliff Grit
16
15
14
13
12
10
9
8
7
6
5
3
2
1
96
94
92
90
88
86
84
82
80
78
76
74
72
70
68
m
Zone Markers
Occurrence found in the outcrop
Uncertain boundary
Stratigraphic gap
Oolitic limestone
Nodular limestones
Marl
Sandstone with sandy limestone
Polarity
Normal polarity
Reverse polarity
Indeterminate polarity
Single sample polarity
Sampling gap
Data Utilization
(codes in DII columns)
Used for polarity interpretation and calculation of paleopoles
Used only for polarity interpretation
Indeterminate polarity, not applicable

numbering of the main magnetozones in this composite is a downward continuation of the "BB" (for British Boreal) nomenclature used by Przybylski et al. (in press-a) for the Oxfordian–Kimmeridgian boundary interval.

Within this integrated magneto-biostratigraphy for the Oxfordian of England and Scotland, a few observations and caveats:

(1) Callovian–Oxfordian boundary interval. The primary source for the polarity pattern spanning the Callovian–Oxfordian boundary is the expanded section and leading GSSP candidate to the east of Ham Cliff (Magnetozones BB21-19). This boundary succession is probably condensed or absent in the other regions. The base of the Oxfordian, as proposed at this GSSP candidate, coincides with a normal polarity on Dorset and the Isle of Skye (the younger part of Magnetozone BB21). The magnetostratigraphy of the underlying *Q. lamberti* Subzone of the *Q. lamberti* Zone and overlying *Q. mariae* Zone is from the Isle of Skye, but with the inclusion of a potential normal-polarity zone in mid- *Q. lamberti* Subzone observed in Yorkshire (CN section).

(2) Lowest Oxfordian. The polarity of *Cardioceras praecordatum* Subzone of the *Q. mariae* Zone (Magnetozones BB18-BB16) is not well resolved. This subzone is condensed in the reference section in Isle of Skye and only partially present in Yorkshire (CN).

(3) Lower Oxfordian. Within the constraints of the high-resolution biostratigraphy and facies changes, the patterns through the *Cardioceras cordatum* Zone (upper Magnetozone BB16 through lower BB14) and within the lower *Perisphinctes pumilus* Subboreal Zone (Magnetozones BB12 through lower BB10) are consistent among all sections. The intervening *Perisphinctes plicatilis* Subboreal Zone is mainly resolved in Dorset (RC, HC and BP sections; normal part of Magnetozone BB14 through the reverse part of BB12).

(4) Middle Oxfordian. The uppermost *P. pumilus* and lowermost *Perisphinctes cautisnigrae* Subboreal Zones were sampled only in Dorset (Black Head), where they are bounded by major unconformities. The upper *P. cautisnigrae* Zone is absent at a major hiatus throughout Dorset, and has not yet been sampled elsewhere.

7. Discussion

7.1. Correlation of the British and Sub-Mediterranean polarity scales

The main features within the middle and upper portions of this British magnetic polarity pattern can be correlated to a composite of overlapping and replicate Oxfordian magnetostratigraphic sections from Poland and Spain in the Sub-Mediterranean faunal province (Przybylski and Ogg, 2007; Przybylski, 2008; Przybylski et al., in press-a,b) (Fig. 7). The definition of Oxfordian ammonite subzones and their grouping into zones within the Sub-Mediterranean faunal province has been undergoing continual revision (e.g., Cariou et al., 1997; Gygi, 2000; Głowniak, 2002, 2006). Several sections in Poland, including some of those used for magnetostratigraphy, have incursions of Subboreal or Boreal ammonite taxa. This enables a framework of possible correlations between the subzones of the different faunal provinces (e.g., Matyja and Wierzbowski, 2002; Głowniak, 2002). When the two regions (Subboreal, Sub-Mediterranean) have identical ammonite subzones with similar definitions (e.g., within the lower Oxfordian *C. cordatum* Zone), there is an excellent correspondence between main features of the magnetic polarity patterns and similar rates of reversals (Fig. 7).

Within intervals that have less certain biostratigraphic correlations, the same general trends in the polarity patterns are present — e.g., dominance of normal polarity in lower and middle *P. pumilus* zone of middle Oxfordian. Unambiguous one-to-one matching of the frequent reversals and complex pattern in each realm is challenged by a combination of uncertainties in equating ammonite subzones, of the incompleteness of the British or Sub-Mediterranean pattern in some intervals, and of possible mis-interpretations or pervasive overprints of the primary polarity in one of the regions. We have merged the major common features of these two polarity patterns, plus schematically indicate potential brief polarity subzones, to compile a generalized Oxfordian polarity scale (Fig. 8). This combined Oxfordian scale will undoubtedly be modified and enhanced by future high-resolution magnetostratigraphy studies.

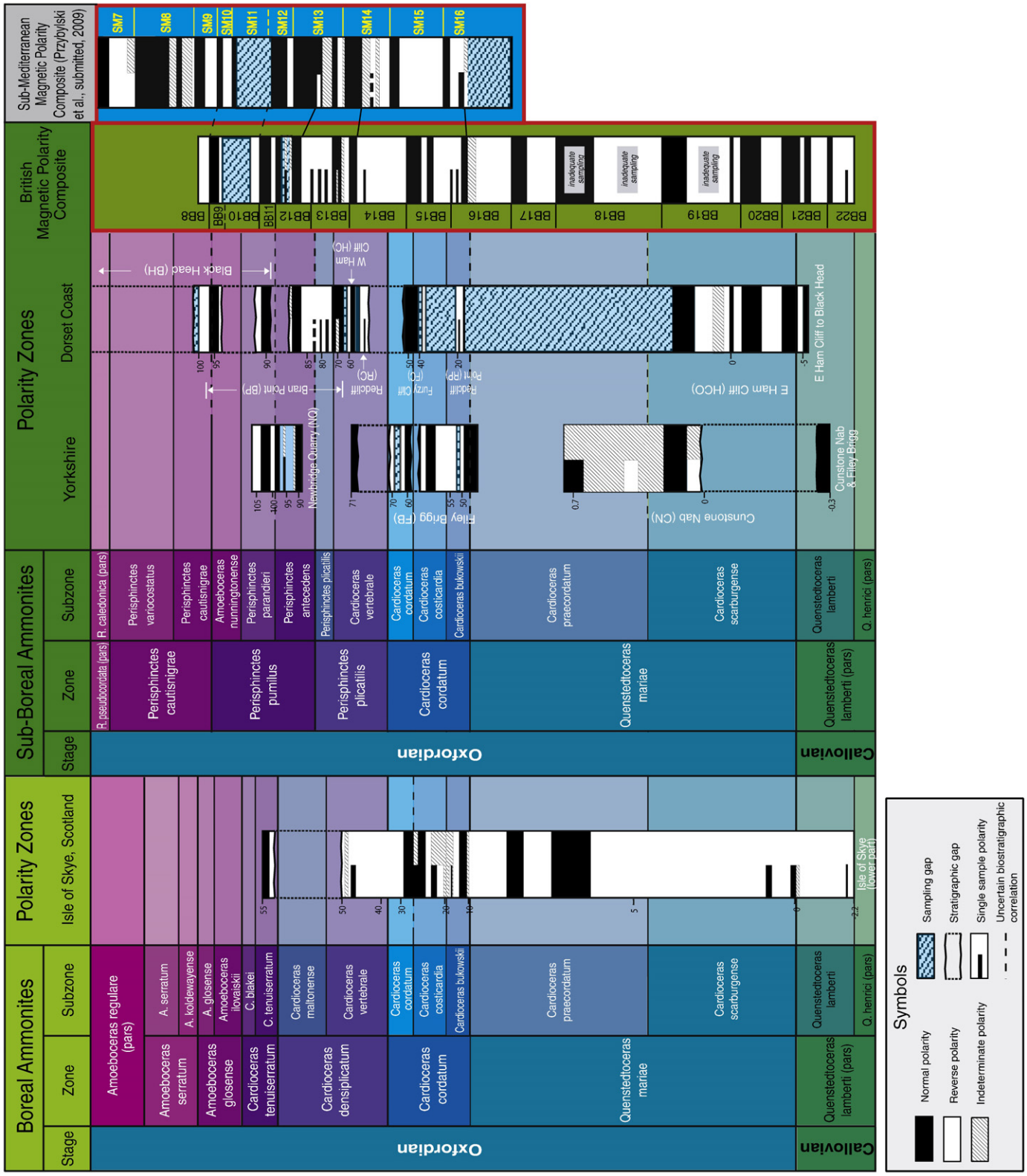
7.2. Calibration of marine magnetic anomalies

There are two geomagnetic models for the pre-M28 portion of the Pacific M-sequence derived from deep-tow magnetometer surveys based on different assumptions of geomagnetic field behavior. The model for the variability observed at 5.5 km depth emphasizes small-scale features that may represent polarity reversals. In contrast, the model based on a mid-depth projection indicates the main features, but may underestimate the number of polarity chrons. The "actual" geomagnetic polarity history is probably between these extremes (Sager et al., 1998; Tominaga et al., 2008). Both versions are displayed in Fig. 8.

The age-scaling of either M-sequence model assumes a constant-spreading rate for the Japanese ridge in the Pacific, in turn calibrated to radiometric-dated ODP sites (e.g., Ogg and Smith, 2004; Tominaga et al., 2008). In contrast, the magnetostratigraphy of Oxfordian sedimentary sections is distorted by variable accumulation rates within and among biozones. It is possible to add a relative age scale and partially remove the longer-term variations in sedimentary accumulation, thereby enabling a more realistic comparison to the M-sequence models, by incorporating estimates of the actual durations of each lower and middle Oxfordian ammonite subzone derived from cycle-stratigraphy interpretations (Boulila et al., 2008, in press; Boulila, 2008). For example, these cycle stratigraphy studies indicate that the *Q. mariae* Zone, which was generally quite thin or condensed in our magnetostratigraphy reference sections, has a relatively long duration nearly equivalent to the next three overlying ammonite zones combined. Unfortunately, these reference sections for Oxfordian cycle stratigraphy in southeast France are unsuitable for magnetostratigraphy (e.g., Kechra et al., 2003; and our own pilot studies).

Even though there are frequent polarity alternations within the pre-M25 extension of the marine magnetic anomaly models, there are also broad trends in the relative dominance of normal versus reversed polarity (Fig. 8). The detailed pattern depends on the selected interpretation of the deep-tow magnetometer data, therefore we concentrated on these broad trends for a first-order correlation to the combined British and Sub-Mediterranean polarity patterns. The correlations are constrained by prior correlations of Kimmeridgian magnetostratigraphy to M24–M24A–M24B–M25 to the Sub–Mediterranean ammonite zones (e.g., Ogg et al., 1984; Ogg and Gutowski, 1996) and of upper Oxfordian magnetostratigraphy to M26 through M30 (Przybylski, 2008; Przybylski et al., in press-a,b). This implies

Fig. 7. Boreal and Subboreal Oxfordian magnetic polarity composite derived from British sections compared with the composite from sections in the Sub-Mediterranean faunal province (Przybylski et al., in press-b). Ammonite subzones in the *C. cordatum* Zone are common between the two faunal provinces, but Middle Oxfordian subzones differ between the provinces. Relative durations of biozones (e.g., the expanded basal Oxfordian *Q. mariae* zone) are scaled according to the cyclostratigraphy study of Boulila (2008) and Boulila et al. (2008, in press).


Boreal Ammonites
Polarity Zones
Isle of Skye, Scotland
Sub-Boreal Ammonites
Polarity Zones
Yorkshire
Dorset Coast
British Magnetic Polarity Composite
Sub-Mediterranean Magnetic Polarity Composite (Przybylski et al., submitted, 2009)
Stage
Zone
Subzone
Oxfordian
Callovian
Amoeboceras regulare (pars)
Amoeboceras serratum
A. serratum
A. koldewayense
Amoeboceras glosense
A. glosense
Amoeboceras ilovaiskii
Cardioceras tenuiserratum
C. blakei
C. tenuiserratum
Cardioceras densiplicatum
Cardioceras maltonense
Cardioceras vertebrale
Cardioceras cordatum
Cardioceras cordatum
Cardioceras costicardia
Cardioceras bukowskii
Queenstedtoceras mariae
Cardioceras praecordatum
Cardioceras scarburgense
Queenstedtoceras lamberti
Q. henrici (pars)
Isle of Skye (lower part)
R. pseudocordata (pars)
Perisphinctes cautisnigrae
R. caledonica (pars)
Perisphinctes variocostatus
Perisphinctes cautisnigrae
Amoeboceras nunningtonense
Perisphinctes pumilus
Perisphinctes parandieri
Perisphinctes antecedens
Perisphinctes plicatilis
Perisphinctes plicatilis
Cardioceras vertebrale
Cardioceras cordatum
Cardioceras costicardia
Cardioceras bukowskii
Cardioceras praecordatum
Cardioceras scarburgense
Quenstedtoceras lamberti
Q. henrici (pars)
Black Head (BH)
Brean Point (BP)
Redcliff (RC)
W Ham Cliff (HC)
Newbridge Quarry (NQ)
Filey Brigg (FB)
Fuzzy Cliff (FC)
Redcliff Point (RP)
Cunstone Nab (CN)
E Ham Cliff (HCO)
E Ham Cliff to Black Head
Cunstone Nab & Filey Brigg
BB8
BB9
BB10
BB11
BB12
BB13
BB14
BB15
BB16
BB17
BB18
BB19
BB20
BB21
BB22
inadequate sampling
SM7
SM8
SM9
SM10
SM11
SM12
SM13
SM14
SM15
SM16
Symbols
Normal polarity
Reverse polarity
Indeterminate polarity
Sampling gap
Stratigraphic gap
Single sample polarity
Uncertain biostratigraphic correlation

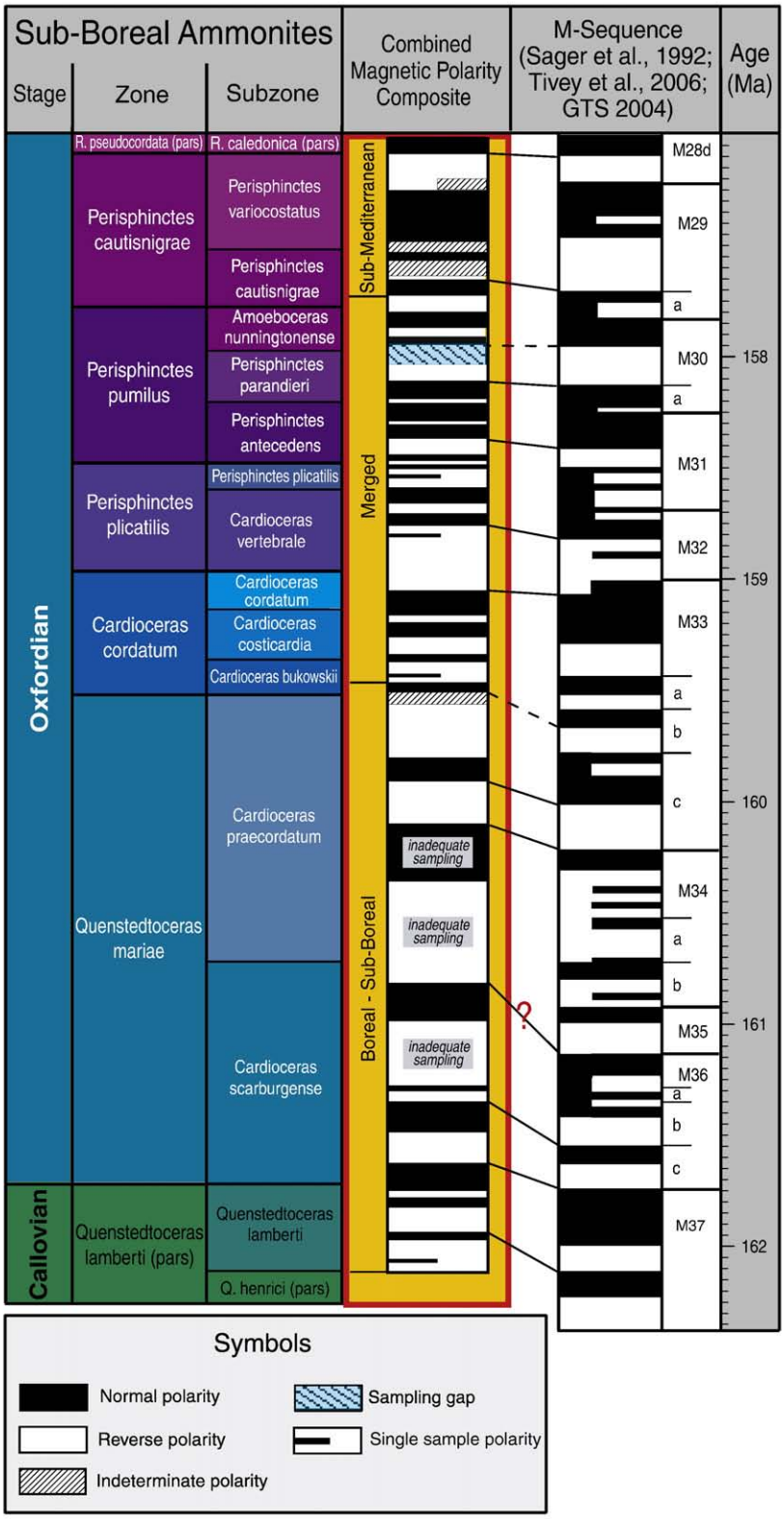


Fig. 8. Correlation of the combined Boreal–Subboreal and Sub-Mediterranean Oxfordian magnetic polarity composites with the M-sequence of marine magnetic anomalies. Relative spans of lower Oxfordian biozones are scaled according to cycle-stratigraphy estimates by Boulila (2008) and Boulila et al. (2008, in press). The right side of the M-sequence column is an interpretation of the deep-tow magnetometer observations at 5.5 km depth; the left side is a mid-depth projection that emphasizes the main features (see text). Projected ages of marine magnetic anomalies based on Pacific spreading rates used in Geologic Time Scale 2004 (GTS2004; Ogg and Smith, 2004) are consistent with ages suggested by Tivey et al. (2006).

that the dominance by reversed polarity at anomaly set M31r–M32r of projected mid-Oxfordian age probably corresponds to the reverse-polarity dominance in the *P. plicatilis* Zone and lowermost *P. antecedens* Subzone. The lack of a distinctive "fingerprint", such as an anomalously longer polarity zone, in either the dual set of M-sequence models or the magnetostratigraphy precludes a precise correlation. Therefore, the proposed matching of the alternating normal- and reversed-polarity intervals assumes that the cycle-adjusted magnetostratigraphy zone

durations are a fairly accurate reflection of the spacing of the marine magnetic anomalies from the constant-spreading rate model (Fig. 8).

The British magnetic polarity scale confirms the basic modeling for the proposed marine magnetic anomalies M30 through M37. Sager et al. (1998) suggested that the polarity pattern "derived from the sea surface model is a more conservative representation of polarity history" because of the "similarity of the short-wavelength anomaly periods to postulated long-term paleointensity variations". The comparison the British composite supports this assessment.

This provisional Oxfordian polarity scale and correlation to either model of pre-M28 marine magnetic anomalies requires future verification. In particular, future work is required to resolve the polarity pattern within the *C. praecordatum* subzone and its presumed equivalence to models for anomalies M33b–M35. This is challenging, because that ammonite subzone is poorly represented in facies that have yielded useful magnetostratigraphy to date.

7.3. Callovian–Oxfordian stage boundary GSSP

Boundaries between units of the geological time scale (GSSPs) are most valuable when established in conjunction with globally isochronous markers. Correlation among the Boreal, Subboreal and Sub-Mediterranean biogeographic provinces of the Late Jurassic and Early Cretaceous is particularly difficult because neither ammonites nor microfossils are sufficiently cosmopolitan to provide adequate datum horizons. Therefore, non-biological correlation of stage boundaries, especially with magnetostratigraphy, is an important criterion for the selection of the international boundary stratotype by the International Stratigraphic Commission (Cowie et al., 1986). Magnetostratigraphy in isolation is generally inadequate to provide an unambiguous correlation, but when merged with biostratigraphic or other constraints, then it provides potential high-resolution global time horizons.

Our magnetostratigraphic sampling included the relatively expanded section and leading GSSP candidate for the Callovian–Oxfordian boundary located east of Ham Cliff (HCO, Dorset). A normal-polarity horizon at the *Q. lamberti–Q. mariae* zonal transition (Magnetozone BB21; Fig. 7) is verified by the presence of a narrow normal-polarity band at this same interval at Staffin Bay (Isle of Skye). After adjusting the magnetostratigraphy records for cycle-scaled durations of ammonite subzones of lowermost Oxfordian, this polarity zone is correlated to the younger normal-polarity interval of the marine anomaly M37 (Fig. 8). Our interpretation can be verified when a reproducible polarity pattern is established for the upper Callovian and a duplicate magnetostratigraphy is obtained for this boundary interval in another basin.

8. Conclusions

- A high-resolution integrated time scale of magnetic polarity and ammonite subzones has been established for the Lower and Middle Oxfordian of Britain. The succession can be partially scaled by cyclostratigraphy (e.g. Boulila et al., 2008, in press) to produce a polarity time scale. This British scale is merged with the middle Oxfordian to lowermost Kimmeridgian polarity pattern that includes calibration of the candidate stratotype (GSSP) for the Oxfordian–Kimmeridgian stage boundary (Przybylski et al., in press-a).
- The mean British Oxfordian paleomagnetic pole derived from combined Dorset and Yorkshire composites is 71.3°N, 172.6°E ($\delta p = 4.2°$, $\delta m = 6.1°$). This pole indicates a minor inclination shallowing relative to Oxfordian paleopoles derived from coeval Sub-Mediterranean outcrops (e.g., Ogg et al., 1991; Przybylski et al., in press-a,b).
- Correlation with the marine magnetic anomaly pattern has confirmed that the Oxfordian is not a 'quiet zone' but that there was a complex succession of many short-duration polarity zones during this span of Earth history. The complete Oxfordian polarity pattern shows similarities to the main polarity dominances and general reversal frequencies of models proposed for marine magnetic anomalies observed from Oxfordian-age oceanic crust, and a tentative correlation of the main polarity features will be tested in future work by both paleomagnetists and marine geophysicists.
- The Callovian–Oxfordian boundary interval at the leading candidate stratotype (GSSP) at Ham Cliff, Dorset, and at the coeval interval at Staffin on the Isle of Skye have good paleomagnetic records. The proposed Callovian–Oxfordian boundary is within a narrow normal-polarity zone that is tentatively correlated with marine magnetic anomaly M37n.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi: 10.1016/j.epsl.2009.11.031.

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